Acute and chronic hepatitis

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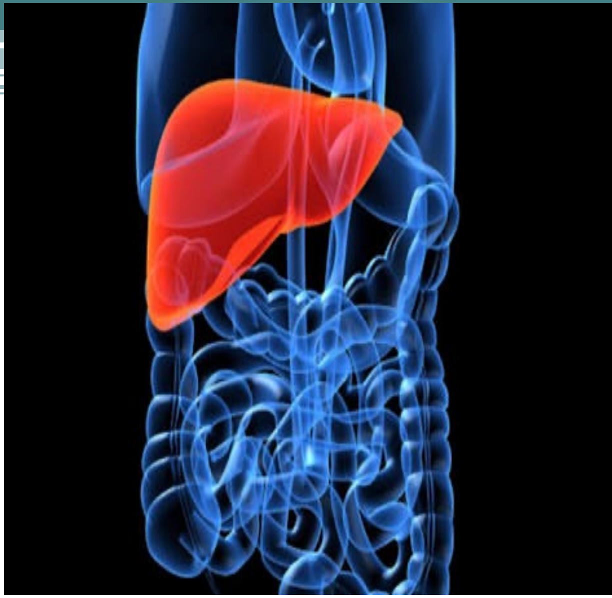


Outline

- Definition
- Classification
- Epidemiology
- Etiology
- Pathogenesis
- Clinical Features
- Diagnosis
- Treatment
- Prevention

Hepatitis

- Inflammation of liver tissue

- Acute <6 months
- Chronic >6 months

- Acute hepatitis:
  - Can resolve on its own
  - Progress to chronic hepatitis
  - Rarely result in acute liver failure

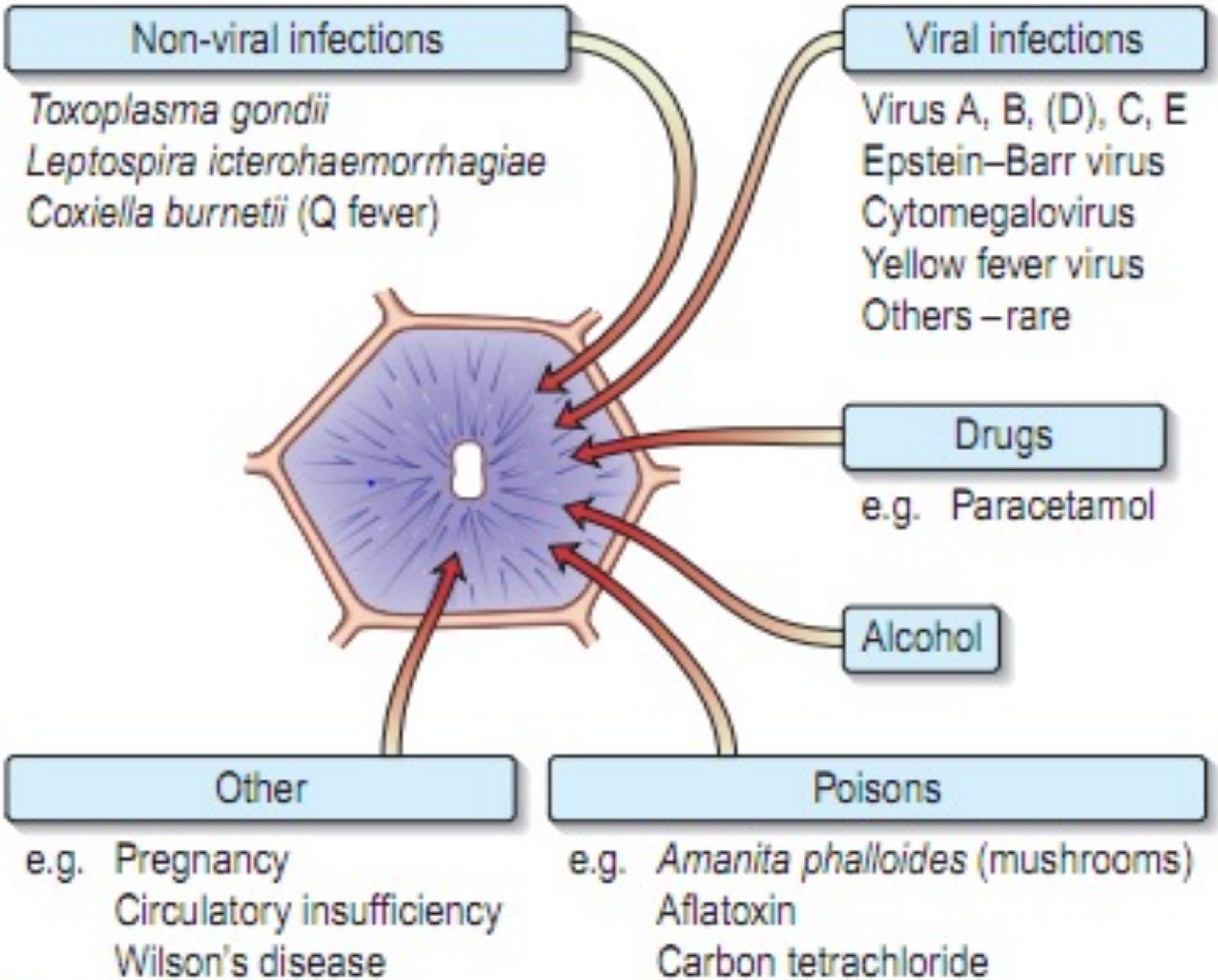


Fig. 7.12 **Some causes of acute parenchymal damage.**

Clinical presentation

- **Three phases**
- **Pre-icteric phase (before jaundice)**
  - Anorexia
  - Fatigue
  - Nausea
  - Vomiting
  - Arthralgia
  - Myalgia
  - Headache
  - Photophobia
  - Pharangitis

- **Icteric phase**
  - ***Jaundice***
  - Enlarged liver
  - Tender upper quadrant
  - Discomfort
  - Splenomegaly (10-20%)
  - Generalized adenopathy

- **Post-icteric phase**
  - Convalescence (recovery)

Lab findings

- Liver enzymes (AST, ALT) increase >5-10 times
- Markers of hepatitis B/C /A/E might be positive

Complications

1. Chronic hepatitis → cirrhosis >> HCC

2. Fulminant hepatitis (Acute liver failure)

Case, DDX

- Acute viral hepatitis
- Infectious mononucleosis
- Toxoplasmosis
- Drug induced hepatitis
- Alcoholic hepatitis
- Cholecystitis/choledocholithiasis

Viral hepatitis

- **Hepatotropic**:
  - Parenteral transmission: B, C, D,
  - Enteral transmission: A, E

- **Non-hepatotropic** :
  - CMV, EBV, HSV, mumps, varicella, yellow fever
  - Mild course without **chronicity**

Hepatitis A Infection

HAV

- RNA virus, picornavirus family

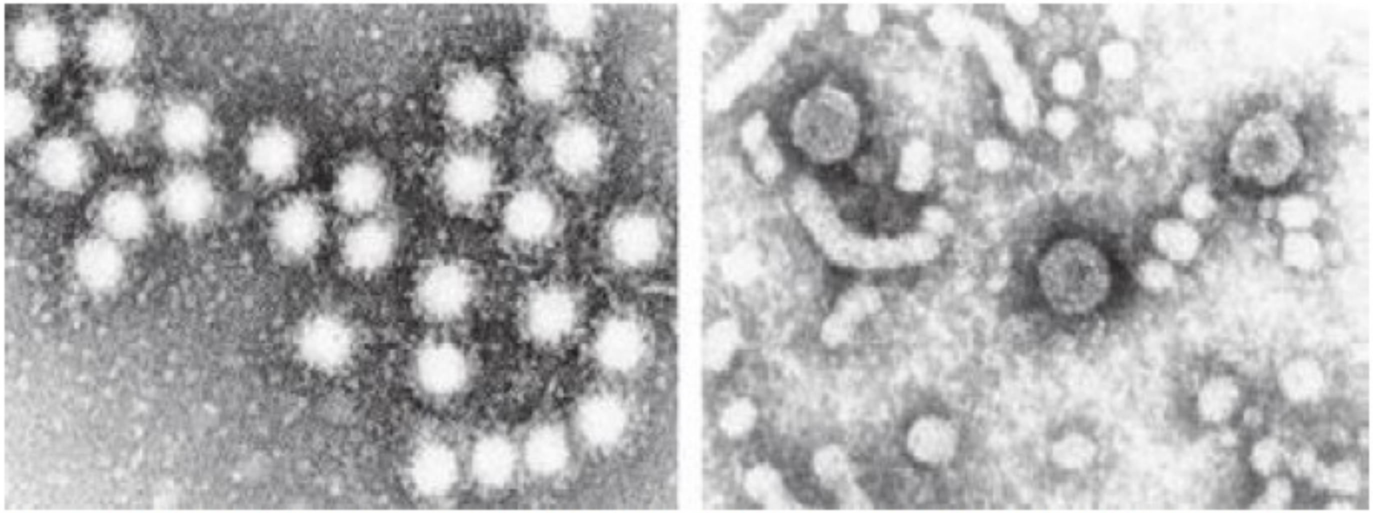


FIGURE 360-1 Electron micrographs of hepatitis A virus particles and serum from a patient with hepatitis B. ***Left:*** 27-nm hepatitis A virus particles purified from stool of a patient with acute hepatitis A and aggregated by antibody to hepatitis A virus. ***Right:*** Concentrated serum from a patient with hepatitis B, demonstrating the 42-nm virions, tubular forms, and spherical 22-nm particles of hepatitis B surface antigen. 132,000×. (Hepatitis D resembles 42-nm virions of hepatitis B but is smaller, 35–37 nm; hepatitis E resembles hepatitis A virus but is slightly larger, 32–34 nm; hepatitis C has been visualized as a 55-nm particle.)

Modes of HAV transmission

- Faeco-oral route (95%)

==> person-to-person contact

==> contaminated food or water

==> salads and fruits washed in contaminated water

==> contaminated shellfish

- Infected plasma (<5%)
- Sexual route (<5%)

HEPATITIS A - CLINICAL FEATURES

•Jaundice by age group:	<6 yrs	<10%
	6-14 yrs	40%-50%
	>14 yrs	70%-80%

•Rare complications:
- Fulminant hepatitis
- Cholestatic hepatitis
- Relapsing hepatitis

•Incubation period:
- Average 30 days
- Range 15-50 days

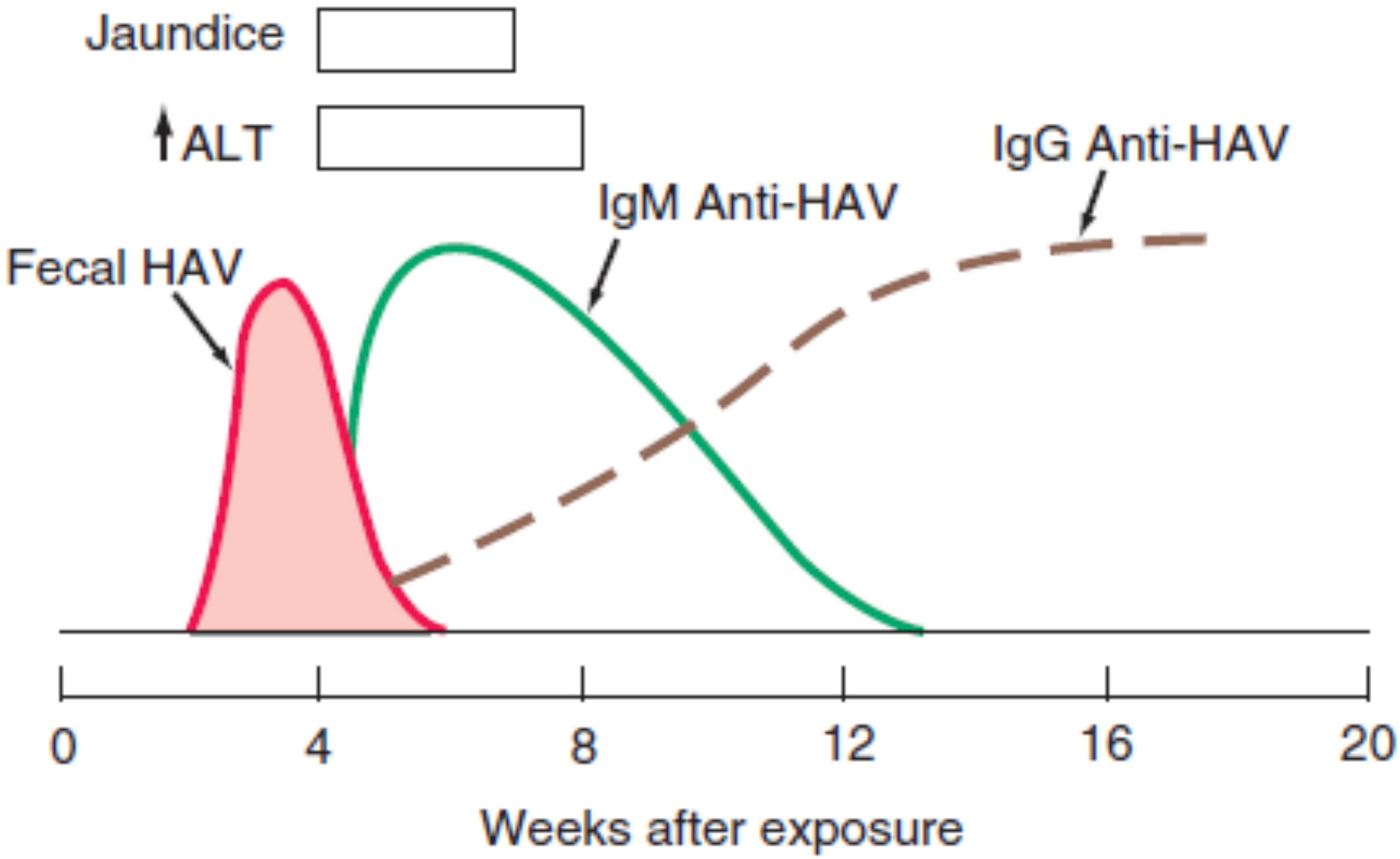


FIGURE 360-2 Scheme of typical clinical and laboratory features of hepatitis A virus (HAV). ALT, alanine aminotransferase.

Investigation

- CBC- leukopnia with relative lymphocytosis
- Liver enzymes and function tests
- IgM anti-HAV indicates acute infection
- IgG indicates previous infection
  - In high prevalence areas most children have Abs by the age of 3 yrs

Treatment and prognosis

- No specific treatment
  - Supportive therapy
  - Admission to hospital not usually necessary
- Excellent prognosis with almost complete recovery
  - Never progresses to chronic liver disease

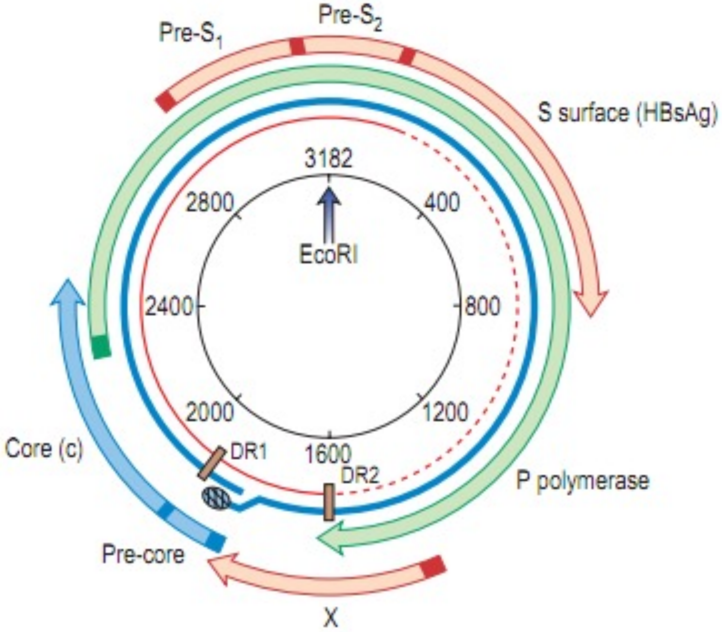
PREVENTING HEPATITIS A

- Hygiene (e.g., hand washing)
- Sanitation (e.g., clean water sources)
- Hepatitis A vaccine (pre-exposure)
- Immune globulin (pre- and post-exposure)

Hepatitis B Infection

HBV

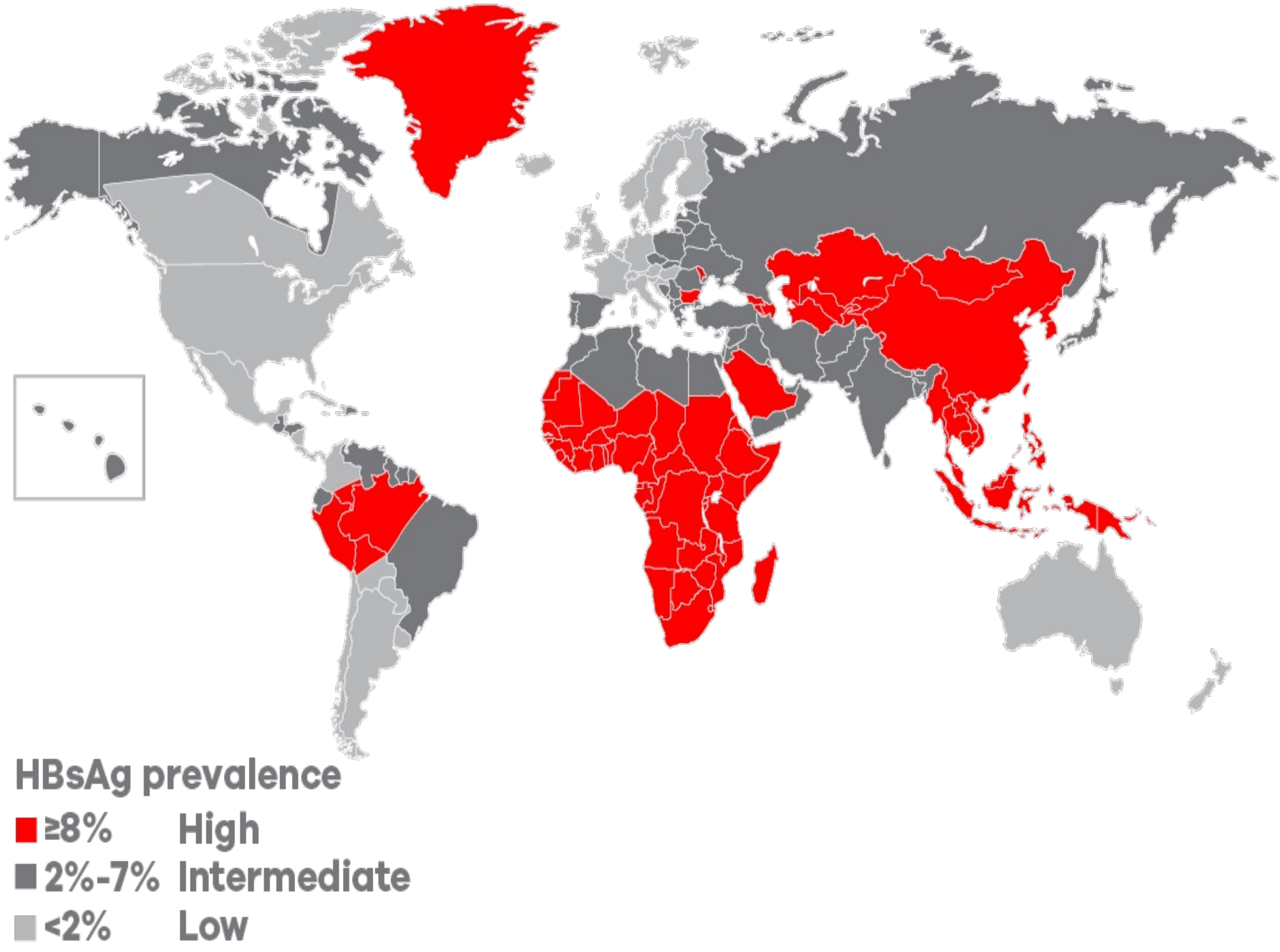
- DNA virus with compact genomic structure
- Family of hepadnaviruses
- 4 overlapping genes- C,S,P,X



HBV, epidemiology

- >2 billion people infected
- 400 million chronic carriers
- 600,000 die each year from complications of CHB

World prevalence of Hepatitis B



Hepatitis B Virus
Modes of Transmission

- Sexual
- Parenteral
- Perinatal

Vertical transmission (Mother to child)

- Commonest world wide

Horizontal transmission

- Esp in children
- Close, household contact
- Commonest in Africa

CDC
CENTERS FOR DISEASE CONTROL AND PREVENTION

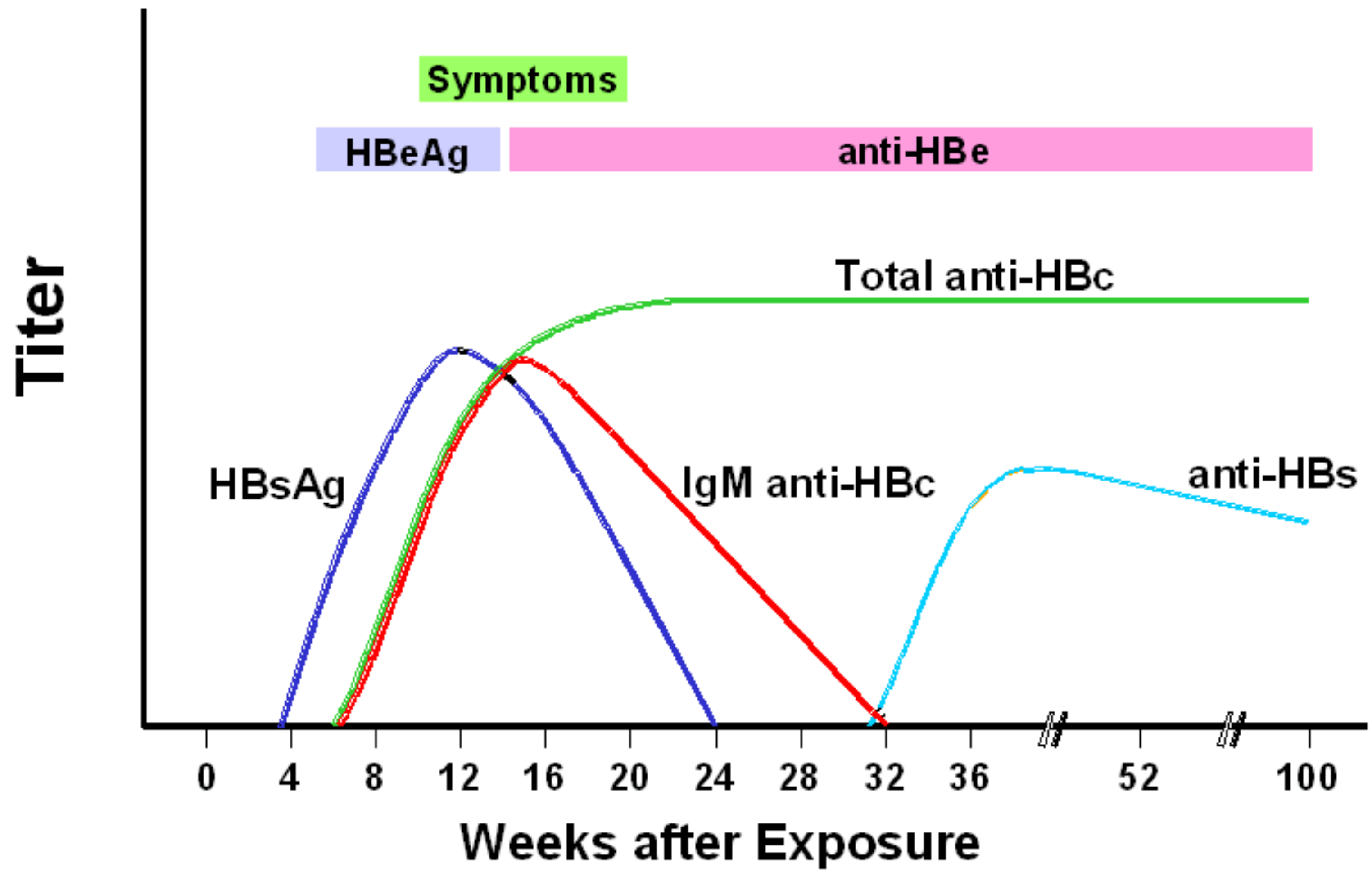
Hepatitis B - Clinical Features

• Incubation period:	Average 60-90 days Range 45-180 days
• Clinical illness (jaundice):	<5 yrs, <10% ≥5 yrs, 30%-50%
• Acute case-fatality rate:	0.5%-1%
• Chronic infection:	<5 yrs, 30%-90% ≥5 yrs, 2%-10%
• Premature mortality from chronic liver disease:	15%-25%

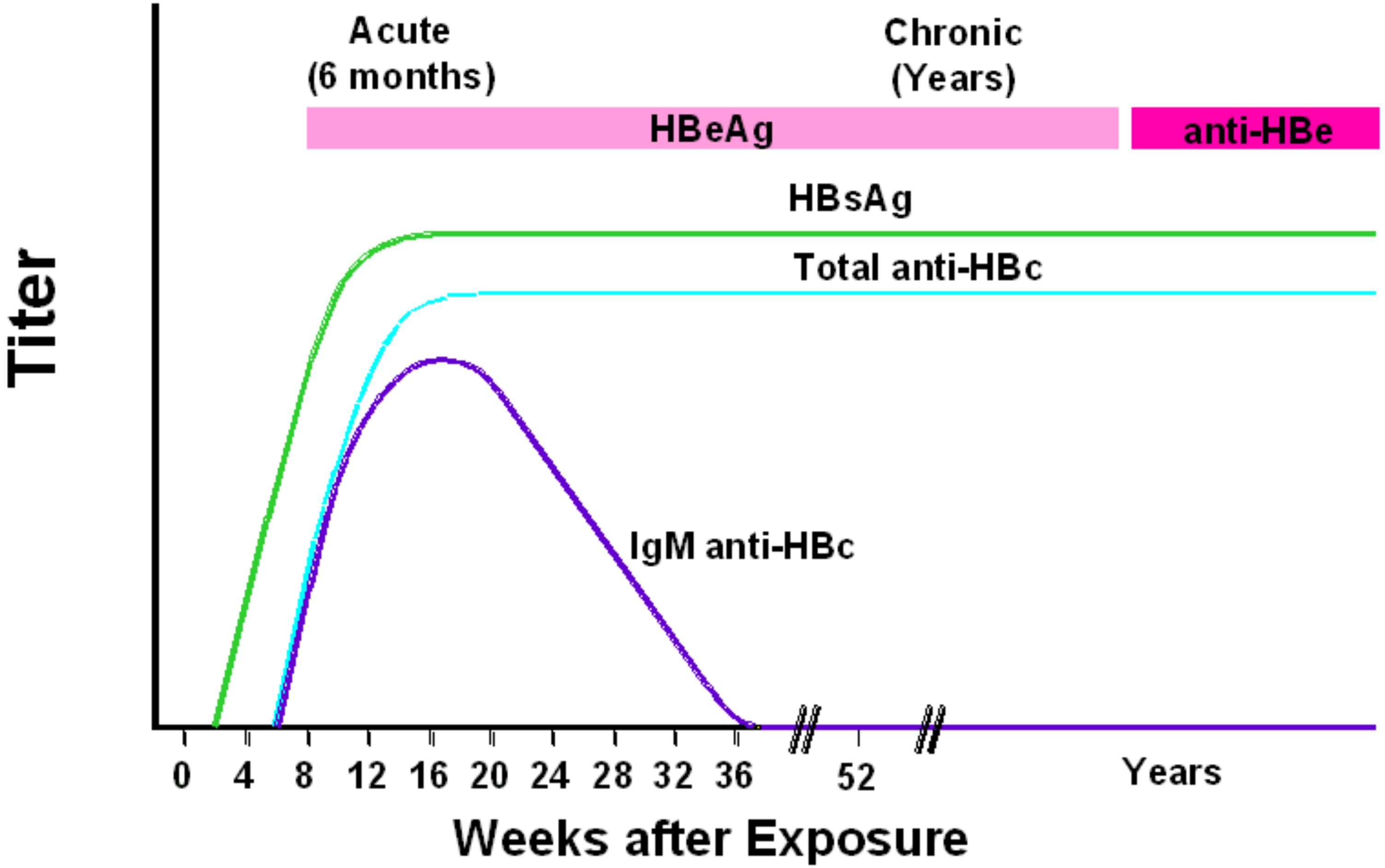
CDC
CENTERS FOR DISEASE CONTROL AND PREVENTION

Table 7.7	Significance of viral markers in hepatitis B
Antigens	
HBsAg	Acute or chronic infection
HBeAg	Acute hepatitis B Persistence implies: continued infectious state development of chronicity increased severity of disease
HBV DNA	Implies viral replication Found in serum and liver
Antibodies	
Anti-HBs	Immunity to HBV; previous exposure; vaccination
Anti-HBe	Seroconversion
Anti-HBc	
IgM	Acute hepatitis B (high titre) Chronic hepatitis B (low titre)
IgG	Past exposure to hepatitis B (HBsAg-negative)

Acute Hepatitis B Virus Infection with Recovery
Typical Serologic Course



Progression to Chronic Hepatitis B Virus Infection
Typical Serologic Course



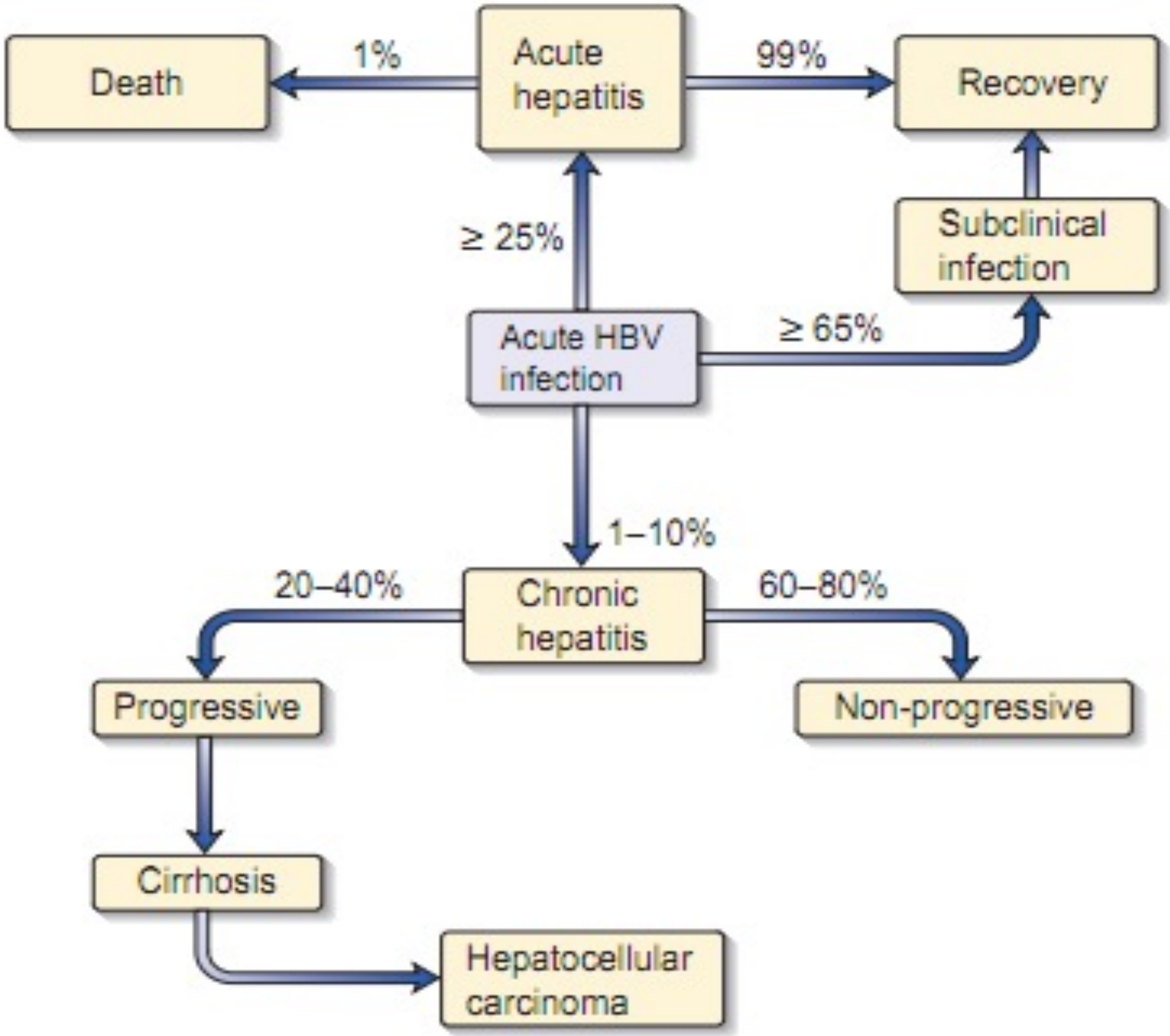


Fig. 7.17 **Clinical course of hepatitis B infection acquired in adults.**

Treatment of acute hepatitis B

- Mainly symptomatic
- Antivirals (TDF,ETV) can be used in very sick patients

PREVENTION STRATEGIES

- Introducing HBV vaccine in EPI program; and
- Mandatory screening of blood donors
- Vaccination of risk groups.
  - Active and passive immunization
- Health education of public and medical personnel.

Hepatitis D

- RNA virus, deltaviridae family
- Common in IVDU
- Unable to replicate on its own, needs HBV
    - Co-infection
        - More severe course
    - Superinfection

- Dx- IgM anti HDV/RNA with evidence of HBV infection
- Prevention? HBV vaccination

Hepatitis C Infection

HCV

- ssRNA, flaviviridae
- 6 genotypes

- 170 million people affected world wide
- Prevalence
  - Southern Europe :1-3%
  - Egypt : 15-20%
  - Ethiopia: 2-3%

Transmission of HCV

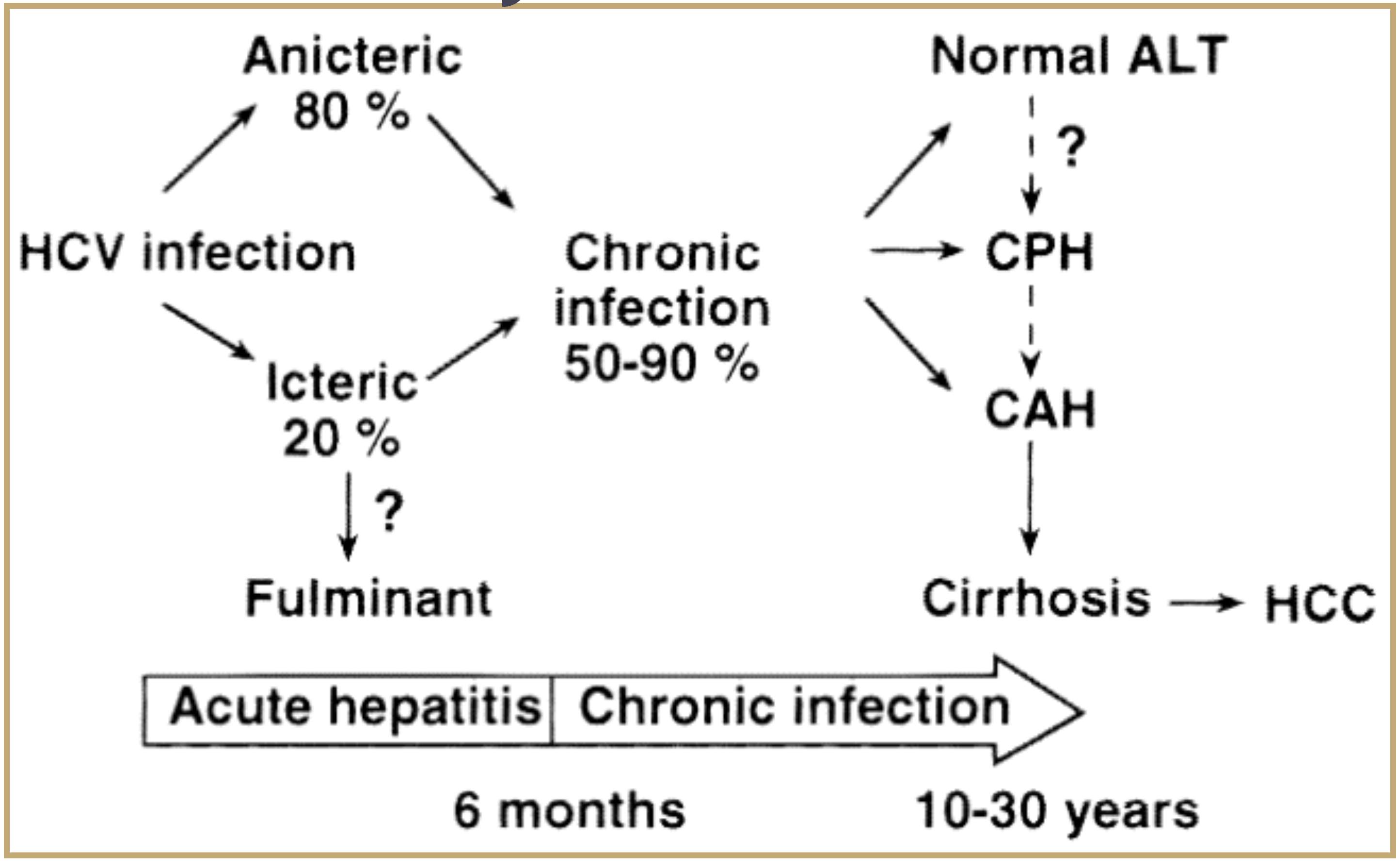
- Percutaneous
  - Injecting drug use
  - Clotting factors before viral inactivation
  - Transfusion, transplant from infected donor
  - Therapeutic (contaminated equipment, unsafe injection practices)
  - Occupational (needlestick)
- Permucosal
  - Perinatal
  - Sexual

CDC
Centers for Disease Control and Prevention

Features of Hepatitis C Virus Infection

Incubation period	Average 6-7 weeks Range 2-26 weeks
Acute illness (jaundice)	Mild (≤20%)
Case fatality rate	Low
Chronic infection	75%-85%
Chronic hepatitis	70% (most asx)
Cirrhosis	10%-20%
Mortality from CLD	1%-5%

Natural history



Marcellin, J Hepat 1999

Diagnosis

- Exclude other viruses
- HCV RNA- detected 1-8 weeks after infection
- HCV antibody – usually after 8 weeks

Treatment of acute hepatitis C

- Controversial
- Needle stick injury with viremia should be treated

Prevention

- No effective vaccine

- Universal precaution
- Health education

Table 7.5 Some features of viral hepatitis					
	A	**B**	**D**	**C**	**E**
Virus	RNA 27 nm	DNA 42 nm	RNA 36 nm (with HBsAg coat)	RNA approx. 50 nm	RNA 27 nm
	Picorna	Hepadna	Deltaviridae	Flavi	Hepevirus
Spread					
Faeco-oral	Yes	No	No	No	Yes
Blood/blood products	Rare	Yes	Yes	Yes	No
Vertical	No	Yes	Rare	Occasional	No
Saliva	Yes	Yes	Yes	? No	?
Sexual	Rare	Yes	Yes (rare)	Uncommon	No
Incubation	Short (2–3 weeks)	Long (1–5 months)	Long	Intermediate	Short
Age	Young	Any	Any	Any	Any
Carrier state	No	Yes	Yes	?	No
Chronic liver disease	No	Yes	Yes	Yes	No
Liver cancer	No	Yes	Rare	Yes	No
Mortality (acute)	<0.5%	<1%		<1%	1–2% (pregnant women10–20%)
Immunization					
Passive	Normal immunoglobulin serum i.m. (0.04–0.06 mL/kg)	Hepatitis B immunoglobulin (HBIg)	No	No	No
Active	Vaccine	Vaccine	HBV vaccine	No	No

Chronic hepatitis

Outline

- Chronic hepatitis
- Chronic viral hepatitis
- Autoimmune hepatitis
- Drug induced chronic hepatitis

Chronic hepatitis

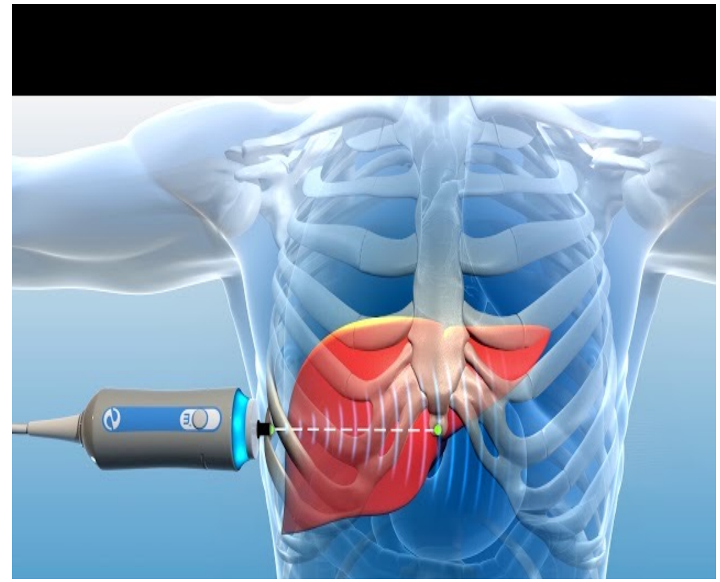
- Series of liver disorders causing hepatic inflammation and necrosis for at least 6 months
- Milder forms are nonprogressive/ slowly progressive
- Severe forms lead to scarring and cirrhosis

CAUSES OF CHRONIC HEPATITIS

- 1. <u>Viral (HBV, HCV, HDV)</u>
- 2. Drugs (alpha-methyldopa, isoniazid)
- 3. Alcoholic liver disease
- 4. Non-alcoholic steatohepatitis
- 5. Metabolic causes
  - a. Primary biliary cirrhosis
  - b. Sclerosing cholangitis
  - c. Alpha-1-antitrypsin deficiency
  - d. Wilson's disease
  - e. Haemochromatosis
- 6. Autoimmune hepatitis
  - a. Type I (antiactin/lupoid)
  - b. Type II (anti-liver kidney microsomal)
  - c. Type III (anti-soluble liver antigen)
- 7. Cryptogenic

- Classification of chronic hepatitis is based on
  1. Its cause
  2. Histologic activity/Grade
     - Degree of necroinflammatory activity
  3. Degree of progression/Stage
     - Assessment of fibrosis (fibroscan, liver biopsy)

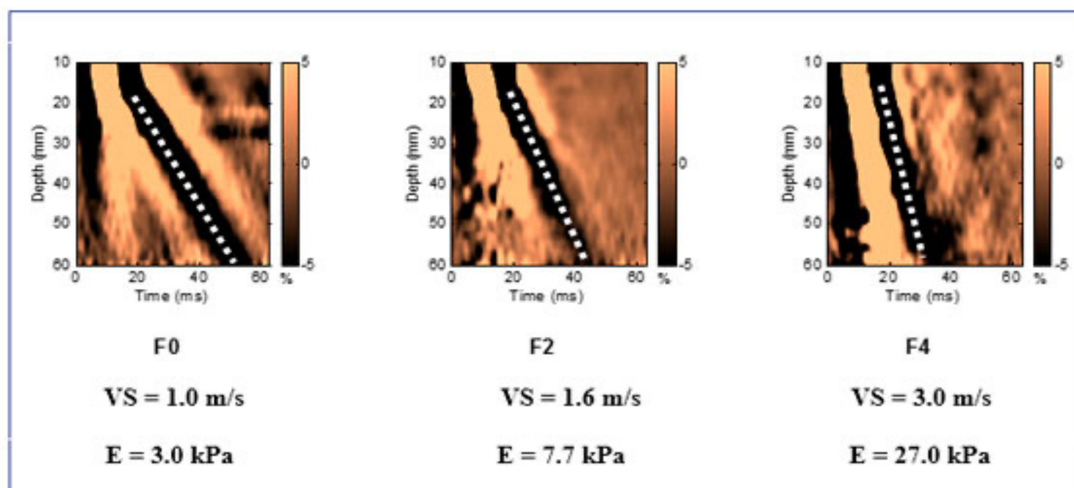
Fibroscan (elastography)



Liver Bx: 1/50,000 of the liver

FibroScan: 1/500 of the liver

Velocità di propagazione dell'onda elastica nel fegato a seconda del grado di fibrosi



Chronic Viral Hepatitis
(B,C,D)

Chronic hepatitis B

- Likelyhood of chronicity after acute hepatitis depends on age
  - Neonates: 90%
  - Immunocompetent adults: 1-5%

- Degree of liver injury is variable
  - None in inactive carriers to mild/moderate/severe
- Inactive carrier
  - HBsAg+, HBeAg-, normal liver enzymes, low or undetectable viremia, normal liver biopsy

- *Viral load correlates with degree of injury and progression*

Clinical feature

- Fatigue – most common symptom
- Jaundice (persistent/intermittent)- advanced
- Complications of cirrhosis
- HCC – can occur with out cirrhosis

Laboratory feature

- Normal to modest enzyme elevation (100-1000), ALT> AST
- Moderate elevation in bilirubin (3-10mg/dl)
- ALP- normal or slightly elevated
- In severe cases
  - Low albumin, prolonged INR
- Markers of chronic hep B
- Viral load

Treatment

- Goals
  - Suppression of HBV DNA to undetectable level
  - Reduce risk of progression, decompensation and death
- Drugs
  - *TDF/TAF (Tenofovir)*
  - *Entecavir*
  - Telbivudine
  - Lamivudine
  - Adefovir
  - PEG-IFN

Chronic hepatitis D (with HBV)

- Similar clinical and lab feature
- Worse prognosis
- Treated with PEG-IFN/Tenofovir

Chronic hepatitis C

- Most common indication for liver transplant in the US
- Chronicity:
  - 50-80%, regardless of mode or age of acquisition
- Most cases are identified incidentally
- Progresses slowly and insidiously

Clinical feature

- Fatigue- most common symptom
- Jaundice is rare

- Other associated diseases - sicca synd, LP, PCT, T2DM

Laboratory feature

- Similar to CHB
  - *Liver enzymes tend to fluctuate more*
- HCV-antibody is first-line test, if positive do viral load (RNA)
- Viral load and genotype

Treatment

- HCV is curable

- Treatment indication:
  - All pts with hep C infection & detectable HCV RNA

Direct acting antivirals (DAAs)

- Novel oral antivirals
- Given in combination for 12-24 weeks

NS3/4A Protease Inhibitors	NS5A Inhibitors	NS5B Polymerase Inhibitors
Boceprevir	Daclatasvir	Dasabuvir
Glecaprevir	Elbasvir	Sofosbuvir
Grazoprevir	Ledipasvir	
Paritaprevir	Ombitasvir	
Simeprevir	Pibrentasvir	
Telaprevir	Velpatasvir	
Voxilaprevir		

Autoimmune Hepatitis

AIH

- Immune system attacks liver cells
- More common in women
- Associated with other autoimmune diseases
- Can lead to cirrhosis and liver failure
- Cause is unknown
  - Genetics and environment

- May be asymptomatic or present with fatigue/ complications of cirrhosis

- High liver enzymes with lesser ALP & bilirubin
- High serum IgG

- Autoantibodies
  - ANA, ASMA, anti-SLA
  - Anti-LKM1

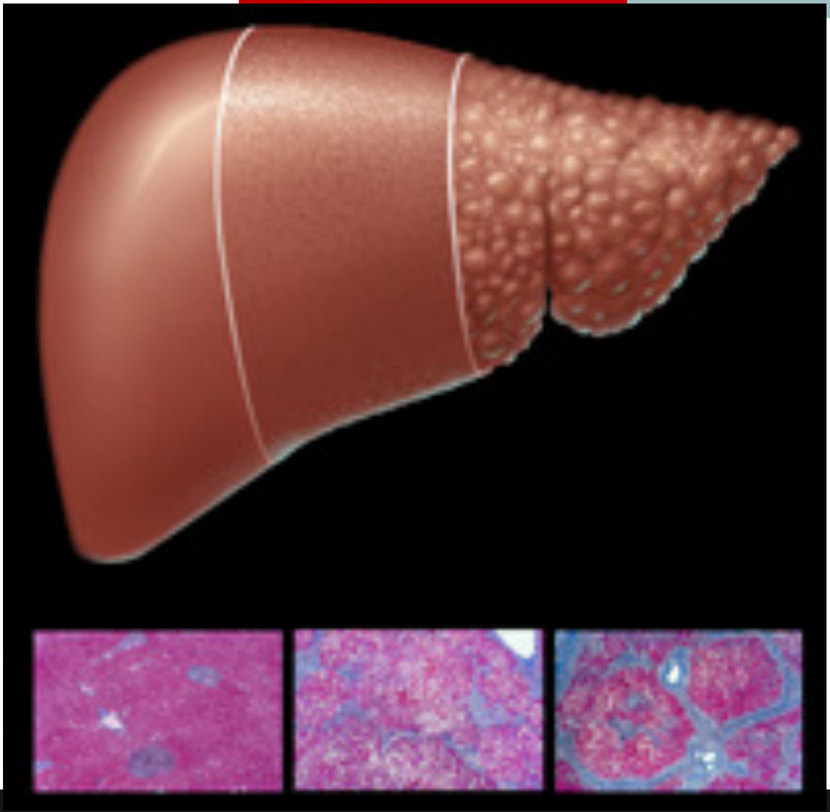
Drug induced chronic hepatitis

- Several drugs can cause chronic hepatitis
- Many have AIH picture
- F>M
- Jaundice, hepatomegaly
- Elevated liver enzymes
- Auto-Abs may be detected
- *Improvement with drug withdrawal and worsening with reintroduction*

- Eg- INH, amiodarone, MTX

Cirrhosis and its complications

Cirrhosis



Definition:

Hepatic necrosis and degeneration combined with hepatic regeneration and fibrosis leading to *nodular formation.*

- Cirrhosis: defined histopathologically
  - Late stage of progressive hepatic fibrosis
  - Distortion of the hepatic architecture
  - Formation of regenerative nodules

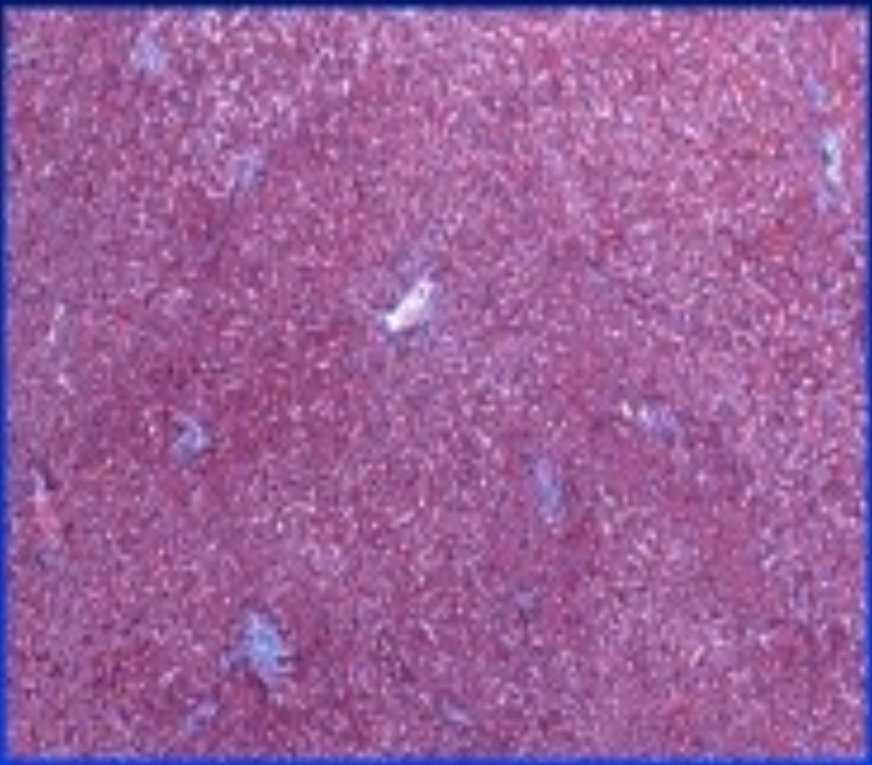
- Final result of chronic liver necro-inflammation
- *Reversible process!*

Normal

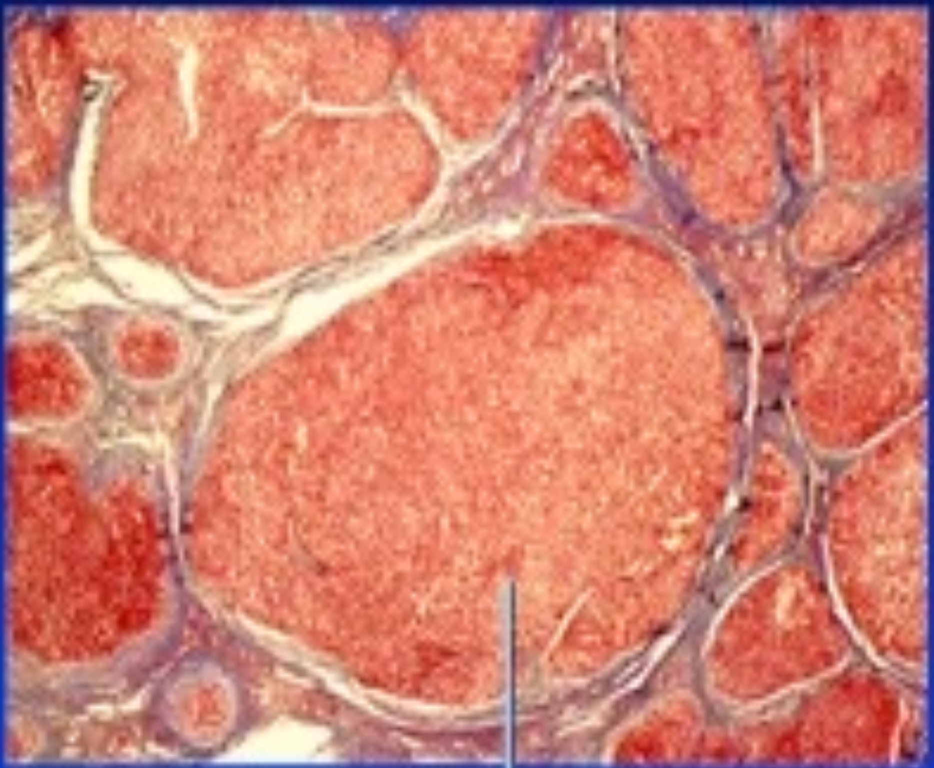
Cirrhosis

Nodules

Normal



Cirrhosis



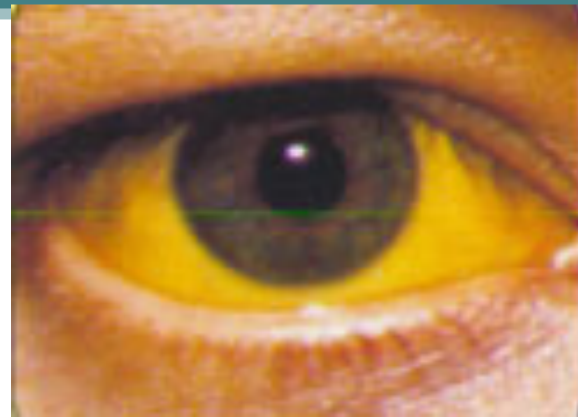
Nodules surrounded by fibrous tissue

Epidemiology

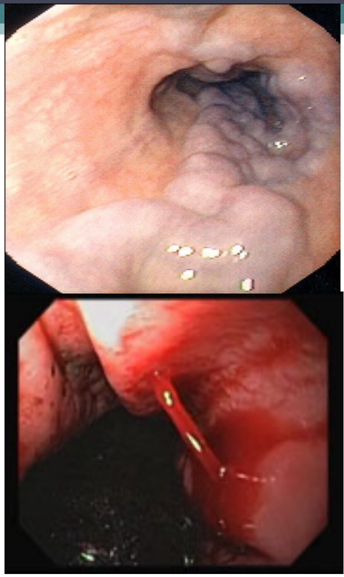
- Cirrhosis affects hundreds of millions world wide
- Prevalence and mortality increasing

Clinical manifestations

- Variable
- 1/3 are asymptomatic
  - May be detected incidentally during laboratory or radiologic testing or an unrelated surgical procedures
- Pts with compensated cirrhosis: anorexia, fatigue, wt loss
- May present with decompensation



Jaundice



Esophageal
Varices

Fluid Retention
(Ascites)

Hepatic
Encephalopathy

Liver Cancer

Clinical Manifestations

Peripheral stigmata

- Spider angiomata
- Palmar erythema
- Gynecomastia
- Testicular atrophy
- Clubbing /hypertrophic osteoarthropathy
- Dupuytren's contracture

Diagnosis

- Liver biopsy
  - Gold standard

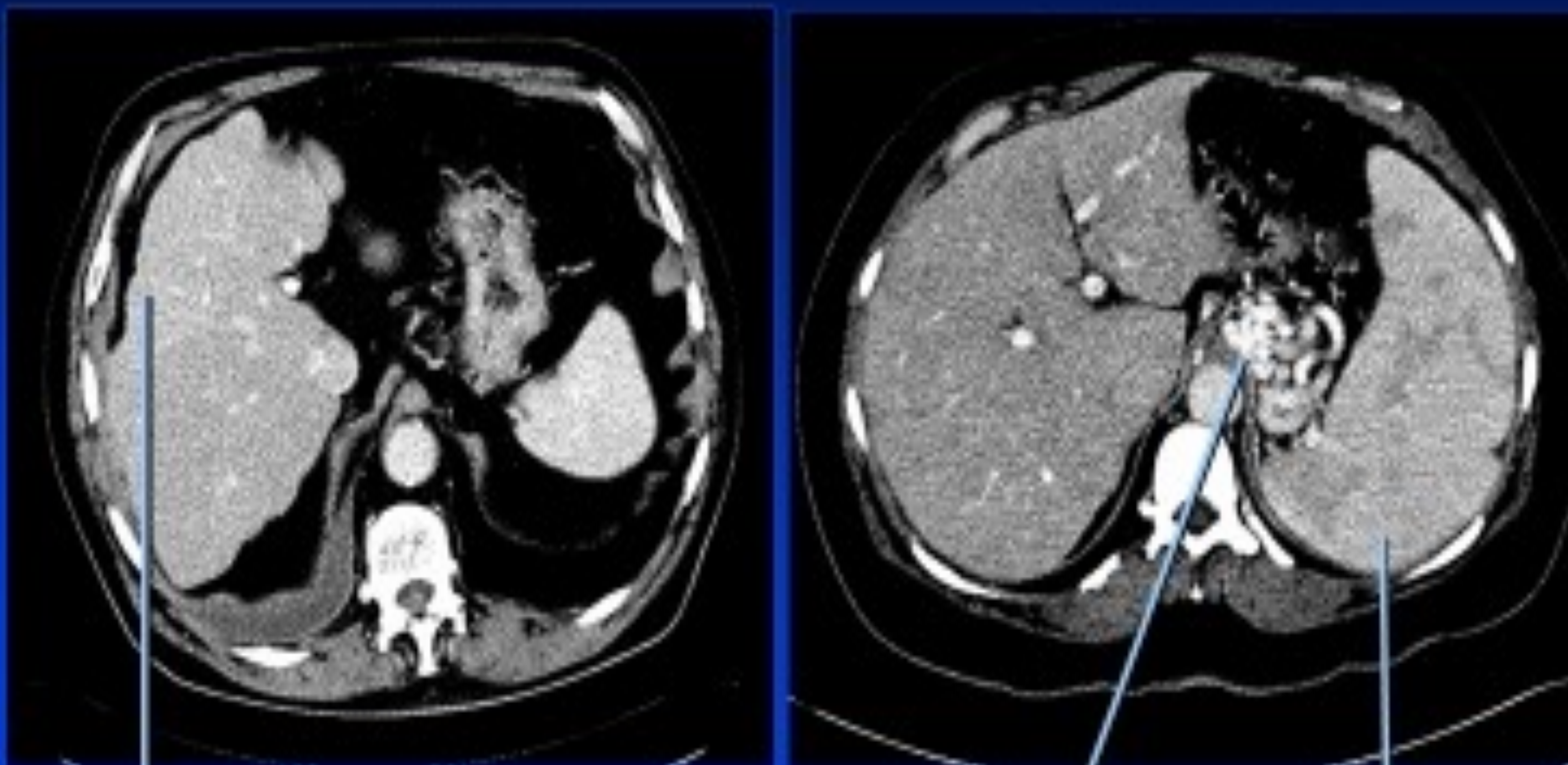
Laboratory findings

- ***Aminotransferases*** - AST and ALT are moderately elevated, usually AST > ALT.
- ***Bilirubin*** - may elevate as cirrhosis progresses
- ***Albumin*** - levels fall with worsening cirrhosis
- ***Prothrombin time/INR*** – prolonged
- ***Thrombocytopenia***

Radiologic investigations

- ***Ultrasound*** is routinely used in the evaluation of cirrhosis
  - Shrunken and nodular liver in advanced cirrhosis

CAT Scan in Cirrhosis



Liver with an irregular surface

Collaterals

Splenomegaly

Natural History of Cirrhosis

- Cirrhosis classified broadly
  - *Compensated*
  - *Decompensated*

- Decompensation:
  - Variceal hemorrhage
  - Ascites
  - Encephalopathy
  - Jaundice

Natural History of Chronic Liver Disease

Chronic liver disease → Compensated cirrhosis → Decompensated cirrhosis → Death

Development of complications:

- Variceal hemorrhage
- Ascites
- Encephalopathy
- Jaundice

Complications of cirrhosis

TABLE 365-2 COMPLICATIONS OF CIRRHOSIS

Portal hypertension	Coagulopathy
Gastroesophageal varices	Factor deficiency
Portal hypertensive gastropathy	Fibrinolysis
Splenomegaly, hypersplenism	Thrombocytopenia
Ascites	Bone disease
Spontaneous bacterial peritonitis	Osteopenia
Hepatorenal syndrome	Osteoporosis
Type 1	Osteomalacia
Type 2	Hematologic abnormalities
Hepatic encephalopathy	Anemia
Hepatopulmonary syndrome	Hemolysis
Portopulmonary hypertension	Thrombocytopenia
Malnutrition	Neutropenia